

Quality Assurance of a Diagnostic X-ray Machine



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1 Objective

2 Apparatus

3 Theory

4 Observation

Listing 1: PDD analysis code

```
1 # Importing some useful library
2 import matplotlib.pyplot as plt
3 import numpy as np
4 import pandas as pd
5 import scienceplots
6 plt.style.use(["science", "notebook", "grid"])
7
8 # Data Extraction from the xlsx file
9 df = pd.read_excel("PDD and Profile/Program/PDD & PROFILE.xlsx", sheet_name=1)
10 x = np.array(df["Depth"])
11 y = np.array(df["Depth dose"])
12
13 # Percentage depth dose
14 PDD = y/max(y)*100
15
16 # To find the surface PD(Percentage dose), dmax, D10(Dose at 10 cm depth), D20
17 # variable initialization
18 SPD = 0 # surface PD
19 dmax = 0; D10 = 0; D20 = 0
20 for i in range(len(PDD)):
21     if PDD[i] == 100:
22         dmax = x[i]
23     if x[i] == 0:
24         SPD = PDD[i]
25     if x[i] == 100: # 100 mm
26         D10 = PDD[i]
27     if x[i] == 200: # 200 mm
28         D20 = PDD[i]
29
30
31 #Plotting
32 plt.figure(figsize=(15, 8))
33 plt.title(r"PDD profile of 10MV SSD=100, Water Phantom, F.S= 10x10cm$^2$",
34         fontsize=20)
35 plt.plot(x, PDD)
36 bbox = dict(boxstyle="round", fc="0.8")
37 plt.scatter([0, dmax, 100, 200], [SPD, 100, D10, D20], color='red', zorder=3)
38 plt.annotate(f"Surface PD: {SPD:.2f}%", xy=(0, SPD), xytext=(25, SPD + 8), bbox=
39     bbox,
40     arrowprops=dict(arrowstyle="->", lw=1.5), fontsize=11)
41 plt.annotate(f"Dmax: {dmax:.2f} mm", xy=(dmax, 100), xytext=(dmax + 20, 95), bbox=
42     bbox,
43     arrowprops=dict(arrowstyle="->", lw=1.5), fontsize=11)
44 plt.annotate(f"PD10: {D10:.2f}%", xy=(100, D10), xytext=(120, D10 + 8), bbox=bbox
45     ,
46     arrowprops=dict(arrowstyle="->", lw=1.5), fontsize=11)
47 plt.annotate(f"PD20: {D20:.2f}%", xy=(200, D20), xytext=(220, D20 + 8), bbox=bbox
48     ,
49     arrowprops=dict(arrowstyle="->", lw=1.5), fontsize=11)
50 plt.ylabel("PDD (%)")
51 plt.xlabel("Depth(mm)")
```

```
47 plt.savefig("PDD and Profile/Program/PDD_10MV.pdf", dpi=300, bbox_inches='tight')  
48 plt.show()
```

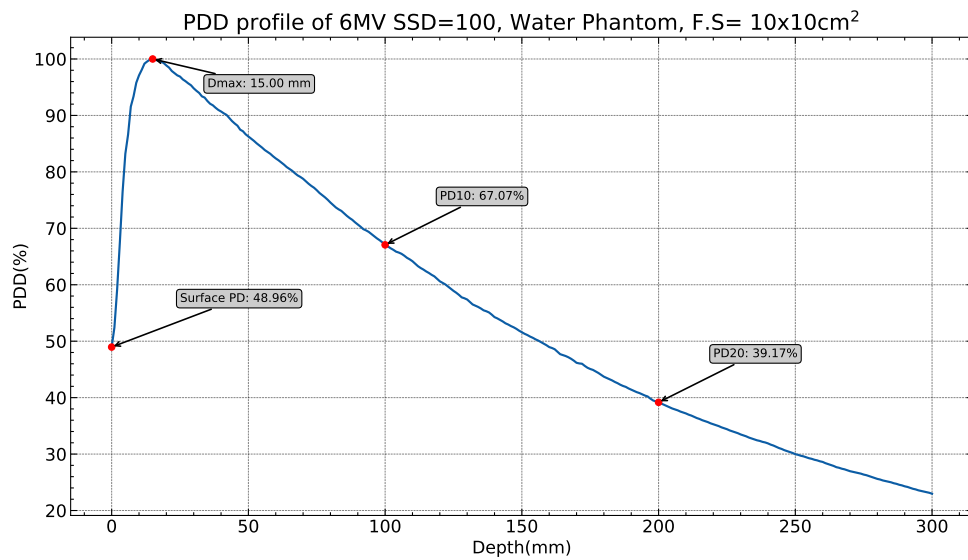


Figure 1: PDD profile of 6MV SSD=100, Water Phantom, F.S= 10x10cm²

5 Conclusion

References